

— SONG YOUNG SHIN DAIRY FARM · REPUBLIC OF KOREA

# Sustainable Dairy Farming

Smart Farming (IT · BT) & Circular Agriculture

스마트팜과 순환농업으로 만드는 지속가능한 낙농

**Xin chào** — 베트남 대표단을 환영합니다 · **Chào mừng đoàn Việt Nam**

**Hyunjae Ha, DVM** 하현제 수의사

Goryeo VET Clinic · Song Young Shin Farm · Farm Study Visit 2026

# A warm welcome to our friends from Vietnam

It is an honor to host public officials of Vietnam at our farm. We share one mission — building a livestock industry that is **profitable for farmers, kind to animals, and gentle on the land**. Today we show how a Korean family dairy turned that mission into daily practice.

회장님의 목장에 오신 베트남 공무원 여러분을 진심으로 환영합니다.

Built around your four areas of interest



## Eco-friendly livestock

친환경축산



## Circular agriculture

순환농업 · Nông nghiệp tuần hoàn



## Sustainable dairy

지속가능한 낙농



## Smart farm

스마트팜 · IT & BT

# What we will walk through today

## 01 Who we are

The D2O ecosystem — farm, clinic, research, platform

## 03 Sustainable dairy

Song Young Shin Farm — welfare, low-carbon, grass-fed A2

## 02 Circular agriculture

순환농업 — the manure-to-soil nutrient loop & D2O

## 04 Smart farming (IT · BT)

smaXtec rumen sensors & genetics / embryo transfer

## 05 A model for Vietnam & Q&A

How this transfers to Vietnamese dairy — and open discussion

# One integrated ecosystem for smart, sustainable dairy



## L&H Research Center

- Sexed embryo transfer (MOET)
- IVF & embryo culture (BT)

## Goryeo VET Clinic

- ET / AI & smart vet system
- Reproduction & feeding consulting

## Song Young Shin Farm

- Animal-welfare & low-carbon certified
- A2 hay milk — Jersey &

## D2O Co., Ltd

- Digital platform for dairy
- Smart-farming solutions (dairy & pig)

# Six forces shaping sustainable agriculture

## Profitability

수익성

A farm must earn to endure.

## Climate change

탄소 · 메탄 · 농업

Low-carbon livestock is now the standard.

## One Health

동물 · 자연 · 인간

Animal, environment & human health are one.

## Smart farming

노동력 · 승계 · 삶의 질

Technology keeps the next generation on the farm.

## Animal welfare

세계적 이슈

Comfortable cows are productive cows.

## Humanity

철학 · 사명감

Farming as a calling, not only a business.

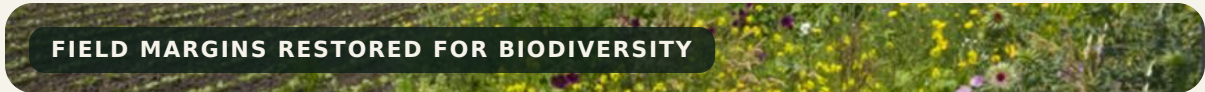
# Waste is not waste — it is a resource in the wrong place

순환농업이란 무엇인가?

- **Manure → Compost → Feed crops → Cattle feed** — a closed nutrient loop
- An eco-economic system that turns a cost into an input
- A direct, practical response to climate change

↓ **Feed costs**

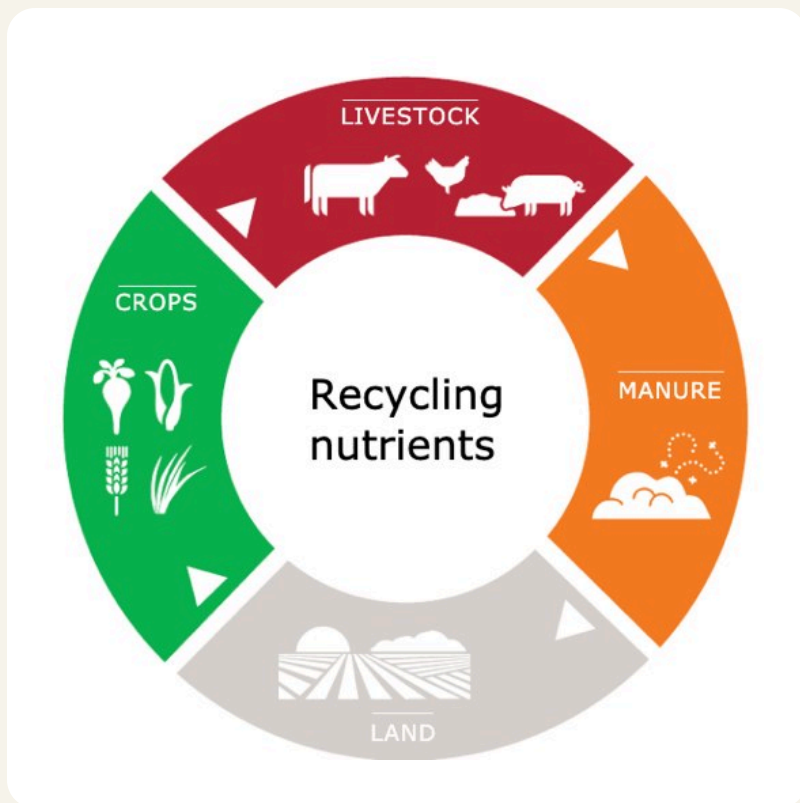
↓ **Pollution & carbon**



FIELD MARGINS RESTORED FOR BIODIVERSITY



# The livestock-crop nutrient cycle



1

## **Feed → Livestock → Manure**

Nutrients circulate instead of leaking out of the system

2

## **Manure = organic fertilizer**

Returned to the grassland that grows the herd's own feed

3

## **Low emission & sustainable**

Fewer inputs bought, less pollution released



## D2O — returning dung to its origin, the soil

The heart of eco-friendly livestock is what you do with manure. D2O is our barn-and-bedding management approach that treats manure as the starting point of the next harvest — not a disposal problem.



### Odor reduction

A cleaner barn for animals, workers and neighbours — critical near communities.



### Healthier bedding

Drier, better-managed bedding supports udder health and animal comfort.



### Low-carbon compost

Faster, controlled composting returns stable organic matter to the grassland.

**Why it matters for Vietnam:** in hot, humid climates manure & odor management is the first bottleneck to scaling dairy near towns. Solve it, and welfare, environment and community acceptance all improve together.



# Song Young Shin Farm

Korea's 1st animal-welfare & low-carbon certified dairy farm

**1<sup>st</sup>** Animal-welfare &  
low-carbon certified

**30<sub>ha</sub>** Own grassland for  
home-grown feed

**A2** Grass-fed hay milk  
Jersey & Holstein



Robotic milking —  
cows choose their time



# No corn. No grain. Just grass.

A sustainable farm produces a distinctive product. Cows grazed and fed on the farm's own grassland give **A2 β-casein milk** — sold as a premium, traceable brand that funds the farm's continued investment in welfare and technology.

- ✓ 100% grass-fed — no corn, no grain
- ✓ A2 Jersey & Holstein hay milk
- ✓ Welfare-first: robotic milking, freedom of movement
- ✓ Farm-to-brand traceability that consumers trust

# Biotechnology: better cows, faster

At L&H Research Center we use **embryo transfer (MOET)** and **IVF** to multiply the best dairy genetics — cows that give more milk, breed reliably, and cope with a changing climate.

- 1 **Oocyte recovery** from selected donor cows
- 2 **Maturation & in-vitro fertilization**
- 3 **Culture** to blastocyst, quality selected
- 4 **Transfer** to recipient — or freeze at  $-196^{\circ}\text{C}$

Focus: **Korean Holstein sexed dairy embryos** — improving milk herds sustainably.



RUMEN BOLUS SENSOR, INSIDE THE COW



— SMART FARMING · IT · SMAXTEC

## One sensor. Three jobs. 24/7.

A small bolus rests in the cow's rumen and continuously measures **inner**  
SMART FARMING · SMAXTEC  
**body temperature ( $\pm 0.01$  °C)**, rumination and drinking — the earliest,

# Catching illness days before you could see it

Inner body temperature is the **earliest health signal** a cow gives. Mastitis and metritis show short fever bouts **2-5 days before** any clinical sign — enough time to act before the animal is sick.

**2-5**

days early warning

**40-70%**

potential savings via early care

## AI — TruAdvice™

The system doesn't just show data — it flags likely conditions:

Mastitis

Ketosis

Milk fever

# Every pregnancy, on time



## Heat detection

- Reads activity, rumination, temperature & drinking together
- Heat index shows how strong the heat is
- Per-cow ideal insemination time — **12-hour window**



## Calving prediction

- Calving alert **15-30 hours in advance**
- Be present for hard births — save cow & calf
- Two calving alerts flag raised **milk-fever risk**



## The result

- Fewer missed heats, shorter calving interval
- Higher reproduction rate = more milk, more calves
- Less hormone use, less guesswork



# Four transition-period diseases, caught early

The riskiest window is right after calving. Each disease has a temperature & rumination fingerprint the sensor reads before the cow looks ill.

## Mastitis

3-5 days early

Several short temperature spikes lasting only hours — before milk yield or conductivity change.

## Metritis / Endometritis

first 14 days

Prolonged fever periods early in lactation, days before clinical symptoms appear.

## Milk fever

around calving

Temperature drop after calving; missed drinking & poor rumination flag cows at risk.

## Ketosis

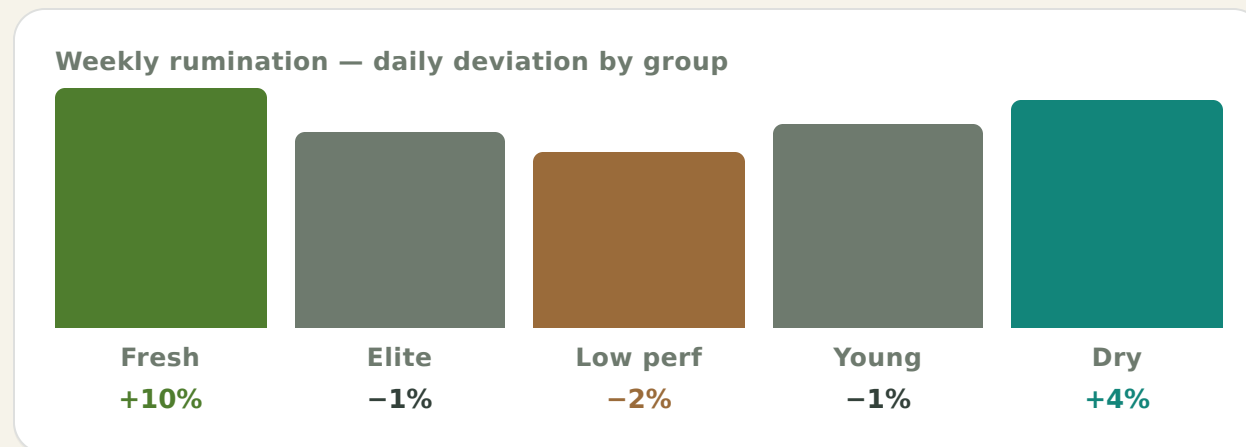
temp + rumination

Temperature & rumination both fall = clinical; often avoided if the earlier disease was caught.

# Feeding, judged by the cow's own rumen

Rumination time is the truest measure of whether a ration is working. The system tracks **weekly rumination and daily deviation for every ration group** — fresh cows, high performers, dry cows, heifers.

- Spot a feeding problem the day it starts, not weeks later
- Balance rations by real digestion, not by eye
- Steady rumen = steady health, fertility & milk



Illustrative of the smaXtec feeding dashboard

# What smart farming delivers

- 1 **Early disease detection** — lower treatment cost
- 2 **Higher reproduction rate** — more productivity
- 3 **Labor saving** — 24/7 automatic monitoring
- 4 **Data-driven decisions** — scientific management

## Fit for Vietnamese farms 🇻🇳

- ✓ Suited to 10-50 head family farms
- ✓ Works in hot & humid climates
- ✓ Simple smartphone monitoring
- ✓ Easy bolus installation, lifetime use per bolus
- ✓ Long-term return on investment

# From this farm to Vietnamese dairy — a shared path

01

## Pilot

A demonstration herd with sensors, welfare housing and circular manure management — proof on Vietnamese soil.

02

## Localize

Adapt to tropical climate, local feed and herd size; train veterinarians & farmers; build the data baseline.

03

## Scale

Extend to cooperatives and provinces — a national platform for eco-friendly, data-driven dairy.

We offer **knowledge exchange, training and technology partnership** — bringing eco-friendly livestock, circular agriculture, sustainable dairy and smart-farm practice together for Vietnam.

OUR PHILOSOPHY · 농민의 세상은 무궁무진합니다

"The farmer  
holds the key to  
humanity's eternal life  
vault."

— Yun Bong-gil, *The Farmer's Reader*, 1927

건강한 흙

→

행복한 소

→

좋은 우유

Soil to Soul

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# 감사합니다

**Cảm ơn quý vị · Thank you**

It is an honor to share our work with you. May our farms grow together.

**Hyunjae Ha, DVM** 하현재 수의사

Goryeo VET Clinic · L&H Institute of Biotechnology

Song Young Shin Farm · D2O = Dung to Origin